Data centers and nitrate in Iowa public water supplies

A reproducibility audit and analysis of public monitoring data

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Abstract

Background A public research repository reported that data-center construction and operation had essentially zero causal effects on nitrate concentrations in Iowa public water supplies during 2005–2026. We assessed whether its data and computations support that conclusion and developed an observed-data replacement workflow.

Methods We froze repository commit 7e1f0e6d2a3c053a47ed9d90b45e894a9e02066a, inspected the source code and manuscript, re-executed the analysis in an isolated output directory, and tested whether an external nitrate input altered the results. We independently downloaded EPA Six-Year Review 4 nitrate records for 2012–2019, verified the named public water-system identifiers against Iowa's public index, and constructed sample-level censoring intervals and unbalanced monitoring panels.

Results The original code generated all 4,176 quarterly nitrate outcomes and their weather covariates rather than analyzing retrieved measurements. Its headline construction and operation estimates reproduced to reported precision, but supplying an extreme external nitrate record changed no output. The purported Callaway–Sant'Anna standard errors were scaled versions of stacked-regression standard errors. Several spatial counts and staggered estimates differed from the written report. The independently acquired EPA data contained 28,103 nitrate records from 1,649 systems, including 17,011 usable routine community-system samples from 845 systems. Of these community samples, 16,489 (96.9%) had unresolved water-stage labels. Observations covered 9,866 of 27,040 possible system-quarter cells among community systems appearing in the release.

Conclusions The repository's causal null, equivalence, and weather-attribution claims are unsupported. This audit establishes neither harm nor absence of harm from data centers. A valid study requires observed raw and finished water records, independently documented construction and withdrawal dates, physical source and supply-network links, and inference reflecting the number of independent exposed water systems.

Keywords Iowa; nitrate; data centers; reproducibility; public water systems; difference in differences

Introduction

Data-center development raises distinct questions about water demand, water-source changes, treatment capacity, and contaminant concentrations. These questions can overlap operationally without having the same causal answer. During the 2025 Central Iowa Water Works conservation response, the utility documented high source-water nitrate and constraints on producing enough treated water. These records establish a water-quality and capacity problem; they do not identify the contribution of data centers to nitrate concentrations. [1,6]

The audited manuscript combined this policy debate with a statewide difference-in-differences analysis and formal equivalence tests. Its main interpretation was stronger than a failure to reject a treatment effect: it asserted zero impact and attributed nitrate spikes to agricultural flushing rather than cooling demand. Such conclusions require a documented connection between observed samples, actual facility events, and a credible comparison group. Reproducible numerical calculations alone cannot establish those connections.

Our objectives were to verify data provenance, reproduce the submitted computations, identify material methodological errors, and determine what an independently acquired public dataset can support. The resulting manuscript reports an audit and observed-data feasibility analysis. It does not substitute generated observations for a missing causal study.

Audit methods

We obtained all files in the repository's 1-gemini directory through GitHub and retained the commit identifier and file hashes. Materials included two Python scripts, a 70-row facility CSV, the manuscript PDF, a technical report, a media article, and prompt and framing documents. We inspected data-flow paths before executing code. To contain the original scripts' file-copying behavior, we imported their functions and redirected OUTPUT_DIR and WORKSPACE_ROOT into a dedicated forensic directory. Their network-fetch hook was replaced with an empty response because the original functions discard the returned response. The original estimator and data-generation expressions were unchanged. [1]

We regenerated the quarterly and annual panels and all reported analysis outputs. To test actual ingestion, we placed an external CSV in the path the script checks, containing a nitrate value of 987654.321 mg/L for Des Moines. We regenerated the panel and compared SHA-256 hashes. This deliberately implausible value was a software challenge input; it was not a water measurement and was never used in the observed-data analysis.

We separately parsed Iowa DNR's Source Water Protection Tracker index and matched all 48 system IDs used by the original project. We checked selected primary source-water descriptions and a Meta expansion announcement. We also tested the HTTP accessibility of 50 unique URLs embedded in the original manuscript. A successful HTTP response was treated only as link availability, not as validation that the page supports a cited claim. [4,7]

Findings on original data provenance

The function build_pws_nitrate_panels creates a NumPy random generator with seed 20261111. For each of 48 manually described systems and 87 quarters, it computes nitrate values from a hand-entered baseline, seasonal amplitude, source-type sensitivity, manually assigned yearly climate values, and random noise. It then inserts specific values for selected systems and years. Its constructed treatment mechanism adds a small grading pulse and later subtracts an acreage-dependent quantity. Operational status is not given an independently measured nitrate response. These design choices determine the substantive patterns that the subsequent models recover.

The source file includes neither a parser for the returned Water Quality Portal content nor a read_csv call for the purported raw nitrate drop files. File existence triggers a log message, after which the same generator runs. The external-file challenge produced an identical quarterly-panel hash. The 912 rows labeled as cleaned sample data are a selected preview of generated values; they are not the full set of measurements underlying the generated quarterly averages. The fully populated panel therefore does not demonstrate complete historical monitoring coverage.

The original weather covariates also come from a hand-entered year table, quarter offsets, and random perturbations. They are not retrieved PRISM or Iowa Environmental Mesonet measurements. A precipitation coefficient from this system cannot identify the actual causes of the 2025–2026 nitrate events. Neither residualizing on these covariates nor adding fixed effects restores empirical content to generated outcomes.

Table 1 Reported and reexecuted estimates

Quantity	Reported	Reexecuted	Reexecuted SE
Construction TWFE	+0.0013	+0.00129	0.01782
Operation TWFE	−0.0124	−0.01238	0.01420
Claimed group time ATT	−0.0212	−0.01627	0.02082
Stacked static ATT	+0.0018	−0.00747	0.02021

Units are mg/L as N. Every estimate in this table uses generated outcomes and has no evidentiary value for Iowa water quality. The first two reported values are from the manuscript abstract; the group-time value is from manuscript Table 3; the stacked static value is from the Markdown report summary. Displayed group-time SE is the original code's unsupported scaled SE, reproduced only for forensic comparison.

All 48 named PWS identifiers matched the DNR index, and the re-created 70-row facility table matched the repository CSV after CSV parsing. These successful checks validate identities and file reproducibility. They do not validate facility coordinates, project completeness, exact dates, capacities, water-source links, or the nitrate time series. The script's facility CSV contains 20 distinct county names and 6,606.5 MW in summed capacity fields; the narrative's 18 counties and deduplicated capacity require a separate, documented reconciliation.

Findings on design and inference

The purported Callaway–Sant'Anna routine calculates some cohort-time differences, but does not compute the sampling covariance of those estimates. Instead, it sets each dynamic standard error to se_stacked multiplied by 1.04 and the overall standard error to se_stacked multiplied by 1.03. When a cohort-time estimate is unavailable, it substitutes a stacked estimate. These operations are not a valid variance estimator or a principled treatment of unsupported event times. The stacked regression is also not the cohort-by-event interaction-weighted Sun–Abraham estimator named in the manuscript. [8,9]

The cohort-time routine excludes 2006 and 2007 cohorts despite the presence of four such treated systems, including Des Moines and Council Bluffs. Calendar years, rather than the available quarter dates, determine treatment in the quarterly panel, so facilities starting late in a year can be coded as treated from the first quarter. The mechanism variables activate the sum of all linked campus acreage or power at first exposure, assigning later expansions to earlier periods. An operation-only regression after construction has already begun does not isolate the incremental effect of cooling activation.

Table 2 Spatial treatment counts

Definition	Manuscript count	Code count
Same municipality	15	19
Within 5 km	15	21
Within 10 km	21	31
Within 20 km	29	37
Shared or mapped HUC10	24	34

Manuscript counts are from the data section. Code counts are distinct systems flagged as ever treated among the 48 hard-coded systems. These are computational discrepancies, not validation of the exposure definitions.

Municipality matching compares city names, not service-area polygons. Distance calculations use hard-coded system coordinates without evidence that they locate relevant intakes or wellheads. The "upstream" definition is a manually supplied HUC10 adjacency list. Shared hydrologic-unit membership alone does not establish flow direction or a groundwater capture-zone connection. Purchased-water systems also create supply-network dependencies that a proximity or local-watershed cluster may miss.

The TOST formula in the original static-regression routine is recognizable, but a correct formula applied to generated outcomes cannot demonstrate real-world equivalence. The 0.25 mg/L margin is not justified by a clinical or policy loss function; applying the same numeric margin to a binary exceedance outcome instead tests a 25-percentage-point margin. The standardized margin uses the generated outcome SD. The "MDE" is a post hoc approximation based on that SE, not evidence about power in an actual monitoring design.

Finally, a nonsignificant pretrend test does not confirm parallel trends. Few treated clusters, sparse event-time support, common suppliers, and selection into monitoring can all limit the informativeness of such a test. The project's strong claims about parallel trends, mechanism attribution, and a flat zero effect exceed what even an otherwise valid observational DiD would establish. [10,11]

Independent observed data and cleaning

We downloaded EPA's released Six-Year Review 4 chemical-occurrence archive and streamed the SUMMARY_ANALYTE_NITRATE.txt member, retaining STATE_CODE IA and analyte 1040. The archive SHA-256 was 5b8726c9adee8fbfa0ca265393814439d0f5d1a552b7a88b0322d19ec9d738e0. The source covers January 2012 through December 2019; it cannot directly assess events in 2025–2026. We did not append the hybrid-nitrate member because it overlaps nitrate records and includes nitrate/nitrite substitutions. [2,3]

We preserved the original identifiers, result and detection-limit fields, collection dates, source-stage and sampling-point codes. The national text file required cp1252 decoding. The Iowa state filter also returned 17 records with federally administered numeric system IDs, which we preserved rather than forcing into Iowa-prefixed identifiers. Exact duplicate rows were removable, whereas conflicting duplicate analytical IDs would stop the pipeline. No such duplicate IDs were present.

EPA analyte 1040 represents regulatory nitrate; results were interpreted on the nitrate-as-nitrogen basis used for the federal standard. All detected Iowa values were in micrograms per liter and were divided by 1,000. Detection-limit units were converted independently, predominantly from mg/L. For non-detects, we retained the interval from zero to the reported sample-specific limit. The limit divided by the square root of two was used only as a descriptive midpoint convention. We did not assume a common 0.10 mg/L detection limit or treat non-detects as measured zero. Six records lacked usable numeric information. [3,5]

The descriptive community sample was restricted to system type C, routine sample type RT, and usable results or censoring intervals. Source code RW or point codes RW/SR indicated raw water; source code FN or point codes EP/FN/DS/FC indicated finished or entry/distribution water. Contradictory flags would form a separate conflict category. Other records remained unresolved. Thus "finished or entry" is a screening stratum based on available codes, not independent verification that a sample represents an individual customer's tap water.

We calculated within-point quarterly means, then averaged observed sampling points equally within system, quarter, and stage. Annual summaries averaged observed quarters equally and reported quarter counts. Missing periods remained missing. These summaries are neither flow-weighted system concentrations nor population-weighted exposure estimates. Sampling-point maxima were treated as observed maxima, whose interpretation depends on sample frequency. Censoring bounds were retained for concentrations and exceedance indicators.

Table 3 Observed record availability

Measure	Observed count
All Iowa-filtered nitrate records	28,103
Distinct systems in those records	1,649
Community-system records before exclusions	17,017
Usable routine community records	17,011
Distinct usable community systems	845
Finished or entry community samples	521 across 70 systems
Raw community samples	1 from 1 system
Unresolved community samples	16,489 across 829 systems

Observed coverage results

Among the 845 community systems appearing in this release, 9,866 distinct system-quarter cells contained usable routine monitoring observations, or 36.5% of the possible 27,040 cells. The median observed system-year contained one sampled quarter. Only 878 system-years had all four quarters represented; 157 of 6,760 possible system-years had no usable observation. These counts describe this release, not a census of all historical Iowa water systems. The current DNR index contains 1,082 community-system entries, but a different observation period and unknown inclusion rules prevent interpreting that cross-sectional difference as a historical missingness rate.

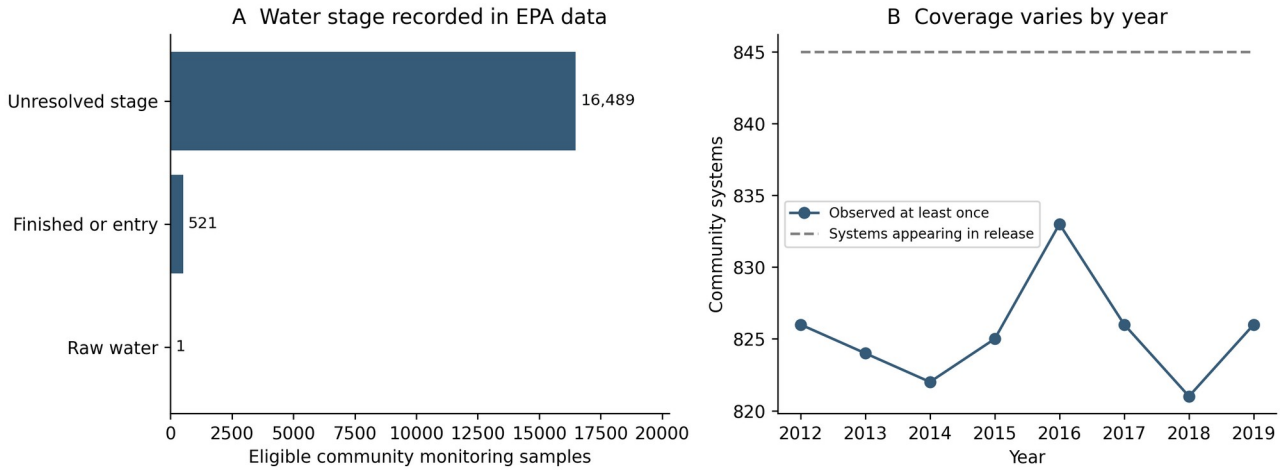


Figure 1 Observed data coverage. Eligible records are routine community-system samples with numeric results or usable censoring limits. Stage is unresolved when neither source nor sampling-point codes establish it. Panel B counts systems with at least one eligible observation in each year. Source EPA Six-Year Review 4.

Table 4 Annual usable monitoring coverage

Year	Samples	Systems	Stage coded samples
2012	2,108	826	37
2013	2,094	824	45
2014	2,056	822	45
2015	2,122	825	50
2016	2,141	833	60
2017	2,159	826	89
2018	2,111	821	87
2019	2,220	826	109

Within the original 48-system list, EPA records were available for 38 systems, totaling 2,355 usable routine samples: 2,322 unresolved, 32 finished or entry, and one raw. Ten named systems had no nitrate record in this release. Absence here does not establish that a utility failed to monitor; it may reflect purchased-water arrangements, release scope, or record availability. The 2020 Water Quality Portal query returned ambient nitrate observations without a direct PWS identifier crosswalk, so those observations were retained separately and not relabeled as public drinking-water samples.

A defensible replacement causal design

The primary causal question should concern the total effect of a verified construction event on nitrate in a physically connected source, with a separately specified finished-water outcome. Raw-water concentration addresses source contamination; finished-water concentration reflects contamination together with treatment, blending, and supplier responses. If demand or treatment changes are on the pathway from development to finished water, controlling for them would remove part of the total effect. They should instead be measured as mechanisms, with direct-effect questions defined separately.

The unit should be an intake, wellfield, or treatment/supply system with stable identifiers and documented changes over time. Distinct customer municipalities purchasing the same produced water should not automatically count as independent exposure units. A longitudinal supplier crosswalk should record purchase shares and dates. Raw and finished samples should remain separate, and sampling-point changes should be audited. Independent source or supply-network clusters should govern uncertainty; local county and HUC clusters can be sensitivity choices when supported by the pathway.

Facility ascertainment must distinguish parcel, campus, building, and project phase, retaining first land disturbance, grading completion, first withdrawal, source of cooling water, wastewater destination, and later expansions. Permit approval is a possible anticipation event, not proof of physical construction or cooling. Missing operation dates must remain unresolved rather than defining an untreated control. Source evidence should include a record identifier, document date, retrieval date, and supporting passage. Year-only dates should define timing intervals and support shifted-date sensitivity analyses.

For surface-water contamination, identify the source catchment and flow path from construction parcels or permitted outfalls to the intake. For groundwater, use aquifer, depth, confinement, and capture-zone information, rather than surface watershed membership. For municipal demand, verify the facility's actual supplier and withdrawal history. Shared HUCs and 5–20 km buffers can be exploratory proximity measures, clearly labeled as such. A city-name match cannot establish any of these pathways.

The supplied replacement estimator calculates transparent paired cohort-time differences against never-treated or not-yet-treated units. For cohort g, it compares the same units at time t and baseline g minus anticipation minus one. Controls must remain unexposed through both periods plus the chosen anticipation interval. It excludes unsupported comparisons and reports cohort sizes, controls, and independent cluster counts. Event-time aggregation uses explicit treated-unit weights; a common-cohort event window should be an additional sensitivity analysis when actual support permits. This is an unadjusted group-time estimator, not a claim to implement the complete doubly robust Callaway–Sant'Anna procedure. [8]

The cluster bootstrap resamples complete supply clusters and recomputes the estimator, instead of scaling another regression's SE. It reports pointwise uncertainty and a maximum-standardized-deviation simultaneous event-study band, subject to support checks. Sparse treated clusters may require an explicitly justified case study or randomization design. Baseline balance, measured local weather, source-specific seasonality, agricultural activity, drought history, sampling intensity, and major treatment changes should be addressed before interpreting parallel trends as plausible. The pipeline currently stops short of a causal estimate because verified exposure links and measured weather were not acquired.

Discussion

The central failure of the original project occurs before econometric model selection: generated values are described as observed longitudinal nitrate measurements. Reproducing the reported headline estimates confirms this data-flow problem rather than the asserted environmental conclusion. Replacing the variance multipliers or choosing a modern DiD estimator would not rescue the original inference. The causal analysis must begin again with auditable measurements and exposure assignment.

Public data access is not wholly blocked. EPA provided a downloadable occurrence archive, Iowa's source-water index supported identifier verification, and a limited Water Quality Portal query returned observations. However, availability does not imply immediate fitness for the proposed causal design. Water-stage ambiguity, irregular sampling, changing source and supplier arrangements, and insufficient treatment timing can make a numerical estimate scientifically misleading. The replacement pipeline preserves these limitations rather than filling the gaps.

A positive contribution of data centers to water demand is compatible with no measurable source-water nitrate effect, and a low finished-water nitrate concentration can coexist with greater treatment costs or reduced capacity. Conversely, an absence of new nitrogen production by cooling equipment does not by itself rule out effects through construction runoff, altered pumping, blending, return flows, or treatment decisions. The relative importance of these pathways requires evidence at the relevant source and supply-network scale. The present audit adjudicates none of them.

Equivalence should be evaluated only after the estimand and inference are defensible. A negligible concentration difference must be justified independently of the observed estimate and should account for baseline concentration and the decision being informed. A symmetric margin for average concentration cannot automatically support equivalence for short-term maxima, exceedance probability, water availability, or treatment cost. Reporting effects with confidence intervals is more informative than declaring a flat zero.

Limitations

This audit concerns a single repository snapshot and does not establish the intent behind its contents. We did not independently validate all 70 facility records, their exact dates, or reported capacities. Our primary observed source ends in 2019 and contains limited stage metadata. The aggregation convention is transparent but is not a validated population exposure measure; censoring substitution remains distribution-dependent, so interval summaries are retained. A link audit checks accessibility, not the truth of all cited claims. Software tests validate selected invariants and implementation behavior, not causal identification.

Conclusion

The repository does not provide empirical evidence that Iowa data centers have zero or negligible effects on public-water nitrate. It also provides no valid evidence of a harmful effect. The accompanying observed-data extract, code, provenance records, and analysis gates provide a reproducible foundation for a new study once source-stage and exposure-linkage evidence are available.

Data and code availability

The accompanying package contains the frozen repository materials, clearly segregated generated forensic outputs, EPA Iowa raw and cleaned records, acquisition hashes, coverage tables, a standalone Colab notebook, Python scripts, and verification checks. No person-level health data were used. Author affiliations, funding, and competing-interest declarations must be supplied by the investigators before submission.

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